

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using a retail transaction dataset containing **3,900 customer purchases** across multiple product categories. The objective is to identify purchasing trends, customer segments, product preferences, and subscription behavior through data analysis and interactive business intelligence.

Tools Used: Python, Pandas, PostgreSQL, SQL, Power BI, DAX

2. Dataset Summary

The dataset contains customer demographic information, purchase details, and shopping behavior.

Dataset Details

- **Total Records:** 3,900
- **Total Columns:** 18
- **Missing Values:** 37 values in the Review Rating column

Key Features

- Customer demographics (Age, Gender, Location, Subscription Status)
- Product information (Item Purchased, Category, Color, Size, Season)
- Purchase details (Purchase Amount, Discounts, Shipping Type)
- Customer behavior (Previous Purchases, Purchase Frequency, Review Rating)

3. Data Preparation and Cleaning (Python)

The dataset was cleaned and transformed using **Python (Pandas)** before importing it into PostgreSQL.

Data Cleaning

- Imported the dataset using Pandas.
- Explored data structure with `info()` and summary statistics with `describe()`.
- Identified missing values in the **Review Rating** column.
- Filled missing ratings using the median review rating of each product category.

- Renamed columns into **snake_case** for consistency.
- Removed redundant columns after validating data consistency.

Feature Engineering

- Created an **Age Group** feature by grouping customer ages.
- Generated purchase frequency information for customer segmentation.

Database Integration

The cleaned dataset was loaded into **PostgreSQL** for SQL-based business analysis.

4. Business Analysis Using SQL

SQL queries were used to answer key business questions and generate actionable insights.

Analyses Performed

1. Revenue comparison between male and female customers.
2. Identification of high-spending customers using discounts.
3. Top 5 highest-rated products based on average review ratings.
4. Comparison of average purchase value by shipping type.
5. Revenue and spending analysis of subscribers vs. non-subscribers.
6. Products with the highest percentage of discounted purchases.
7. Customer segmentation into New, Returning, and Loyal customers.
8. Top 3 best-selling products within each product category.
9. Relationship between repeat purchases and subscription status.
10. Revenue contribution across different age groups.

5. Power BI Dashboard

An interactive dashboard was developed in **Power BI** to visualize customer purchasing behavior and overall business performance.

Dashboard Highlights

- KPI Cards displaying Total Customers, Average Purchase Amount, and Average Review Rating

- Revenue analysis by product category
- Sales distribution across categories
- Revenue and sales comparison by age group
- Customer subscription distribution
- Highest Revenue Products (Top 5 products by Total Revenue)
- Interactive slicers for Gender, Category, Shipping Type, and Subscription Status

DAX Measures Created

- Total Customers
- Average Purchase Amount
- Average Review Rating
- Total Revenue

6. Key Insights

- Clothing generated the highest overall revenue and sales.
- Young Adult customers contributed the largest share of total revenue.
- Express shipping customers had a slightly higher average purchase value than standard shipping customers.
- Loyal customers represented the largest customer segment.
- Subscription customers contributed substantial revenue despite being fewer in number.

7. Business Recommendations

- Introduce loyalty reward programs to improve customer retention.
- Promote exclusive subscription benefits to increase recurring purchases.
- Feature top-rated products in marketing campaigns.
- Optimize discount strategies to maximize profitability.
- Target high-value age groups with personalized promotional offers.

8. Technologies Used

- **Python & Pandas:** Data cleaning and preprocessing

- **PostgreSQL:** Data storage and SQL analysis
- **SQL:** Business query development
- **Power BI:** Interactive dashboard development
- **DAX:** KPI and measure creation

9. Project Outcome

This project demonstrates an end-to-end **Data Analytics workflow**, including data cleaning with Python, SQL-based business analysis in PostgreSQL, DAX measure creation, and the development of an interactive Power BI dashboard featuring KPI cards, customer segmentation, revenue analysis, and Highest Revenue Products for business decision-making.